Black Holes and Time-Mass Duality:

Covariance of G, Evaporation Sequence, and the Neutrino as Final State

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Contents

Status markers

- **[K]** *Confirmed* — algebraically complete.
- **[B]** *Established* — secured by verification script.
- **[S]** *Speculative* — conjecture, still open.

Note. This document is **[S]** as a whole. It draws consequences from $T \cdot m = 1$ (Doc. 003) and $G = \xi^2/(4m)$ (Doc. 012, R58) for a regime — black holes, Hawking evaporation — that the corpus has so far touched only marginally (Doc. 025: Hawking radiation as return to the ξ-field; Doc. 049: field-node tunnelling through the horizon; Doc. 306: horizon as extremum $T_0 = \xi t_P$, $E_{\text{max}} = E_P/\xi$). The algebraic relations are secured by `pruef_350`; their physical validity in this regime is not.

Abstract

From $T \cdot m = 1$ and $G = \xi^2/(4m)$ follows $G \propto T$: an object's gravitational coupling grows with its intrinsic time, while $m \cdot G = \xi^2/4$ stays invariant **[B]**. For an evaporating black hole this means: the sequence of emitted patterns is fixed by the mass hierarchy, and the last stable pattern is the neutrino — the state with the longest intrinsic time and, algebraically, the vacuum of the Clifford-Fock space (Su[0]=Vss[0], Doc. 344) **[B]**. A point-like singularity with finite G is excluded **[B]** (Doc. 078). Whether the electron pattern (or any pattern) is still the same object under strongly altered T remains open **[S]**.

1 Starting point: the question

Solving the FFGFT gravity equation for a Planck-scale mass — does G go to zero while mass remains finite? And is there a calculable difference from the standard calculation?

The answer is not yes or no. The question presupposes that G and m can be varied independently. In FFGFT they cannot.

2 Theorem A — Covariance of G under $T \cdot m = 1$

Theorem 2.1 (G scales with intrinsic time **[B]**). *From $T \cdot m = 1$ (Doc. 003) and $G = \xi^2/(4m)$ (Doc. 012):*

$$G = \frac{\xi^2}{4} T, \qquad m \cdot G = \frac{\xi^2}{4} = \text{const.}$$

Proof. Substitute $m = 1/T$ into $G = \xi^2/(4m)$. □

Reading. If an object's intrinsic time grows (e.g. by time dilation near a horizon), its mass drops and its gravitational coupling rises — the product is a Galois number. "$G \to 0$ at finite mass" is thus excluded: $G \to 0$ would mean $T \to 0$, hence $m \to \infty$.

3 Theorem B — No point-like singularity

Theorem 3.1 (Mass singularity excluded **[B]**). *Since $m \cdot G = \xi^2/4$ is invariant, there is no state with $m \to \infty$ at finite G and none with $G \to \infty$ at finite m. The classical notion of singularity (finite mass in vanishing volume at constant G) has no counterpart in FFGFT.*

Proof. Doc. 306 (R50) derives the exclusion natively: from $T \cdot m = 1$ follows $E \to \infty \Leftrightarrow T \to 0$; but FFGFT's time has a minimal, non-vanishing increment $d_1 = 100\,\xi_1 = 1/75$, fixed by $\xi = 4/30000$. Hence T is bounded below and E above. The horizon of a black hole is the *extremum*: slowest time $T = T_0 = \xi t_P$, maximal concentration $E_{\text{max}} = E_P/\xi$, with exact closure $T_0 E_{\text{max}} = 1$ — all finite. Doc. 078 states the same as an exclusion theorem ($m = \infty$ forces $T = 0$, outside the domain). With $G = \xi^2/(4m)$ (Doc. 012) it follows that $G \to 0$ and $G \to \infty$ are excluded as well. □

Note on the derivation. Docs. 025/063 originally justified the exclusion via Heisenberg's energy-time uncertainty; R50 *replaced* that derivation by the corpus-native one (Doc. 306) — the uncertainty relation is not applicable here (no canonical time operator, Pauli). The conclusion (no singular beginning, finite core with ξ-set scale) is unchanged. Consistent with R58 (no separate quantum gravity needed because T^4/Z_3 is pre-ordered). What remains **[S]** is only the application to the interior *beyond* the horizon — the horizon itself is, per Doc. 306, the extremum, not a transition to a singularity.

4 Theorem C — Hierarchy of intrinsic times

Theorem 4.1 (Neutrino as state of longest intrinsic time **[B]**). *Under $T \cdot m = 1$, $T_\nu/T_e = m_e/m_\nu \approx 10^7$ and $T_e/T_t = m_t/m_e \approx 3 \cdot 10^5$. The neutrino is the massive fermion with the longest intrinsic time. Algebraically it is the vacuum state: Su[0]=Vss[0] (Doc. 344, Theorem C) **[B]**.*

Doc. 025 gives the same hierarchy as $\tau_n = \tau_\xi/\xi^n$: each structural level (particle, atom, molecule, cell) has a time scale longer by $1/\xi$. The neutrino sits at the bottom of the mass scale and thus at the top of the time scale.

5 Theorem D — Evaporation sequence

Theorem 5.1 (Order of Hawking emission by mass **[S]**). *An evaporating black hole loses mass; as $M \to 0$, T_{hole} grows. The emittable patterns at each T are those whose own intrinsic time matches T_{hole}. The sequence follows the mass hierarchy:*

$$t \to b \to \tau \to c \to \mu \to s \to d \to u \to e \to \nu.$$

The last stable pattern before complete dissolution is the neutrino.

Corpus connection. Doc. 025 names Hawking radiation as "energy return to the ξ-field". Theorem D sharpens this: the return runs through the mass hierarchy and ends at the state that is algebraically already the vacuum. Doc. 049 describes the same process as field-node tunnelling through the horizon — Theorem D orders the nodes by intrinsic time.

6 What happens to an electron near the horizon [S]

Near the horizon local time is dilated. Under $T \cdot m = 1$: m_e drops, G_e rises. But m_e is not isolated — $\alpha = \xi E_0^2 / K_{\text{frak}}$ (Doc. 011), $v = m_e/((4/3)\xi^{3/2})$ (Doc. 006), G_F, all Yukawa couplings hang on the same scale. If T changes, all change coherently.

Hence "what happens to the electron" cannot be answered without settling whether the electron *pattern* — Galois orbit O_1, QR class, Sd[3] state (Docs. 346–349) — is still the same object under altered T. The orbit is an algebraic invariant; the mass assigned to it is not. Whether the result should still be called "electron" is a matter of definition that the corpus does not decide.

What leaves as Hawking radiation is thermal — a mixed state in which the identity of the infallen pattern survives not as a named particle but as a contribution to the spectrum. The information paradox is not resolved by this, only relocated into the T-structure of the final state **[S]**.

7 Answer to the original question

1. $G \to 0$ at finite mass: **no** — $G \propto T$, and $T \to 0$ forces $m \to \infty$ **[B]**.
2. Calculable difference from the standard calculation: **yes**, but not as a numerical value of G (locally identical, Doc. 012) — as structure: $m \cdot G = \xi^2/4$ is a Galois number, and the Planck scale is the scale at which the *local* values of G, $\hbar$, c fit together — not a universal limit **[S]**.
3. What remains at the end: a neutrino-like pattern, then the vacuum Vss[0] **[S]**.

8 Summary

Table 1: Results Doc. 350

Theorem	Content	Status
A	$G = \frac{\xi^2}{4}T$; $m \cdot G$ invariant	**[B]**
B	No mass singularity; horizon = extremum (Doc. 306/078)	**[B]**
C	Neutrino = longest intrinsic time = vacuum	**[B]**
D	Evaporation sequence ends at the neutrino	**[S]**
—	Identity of the electron under altered T	open **[S]**

Verification script

`pruef_350_schwarzes_loch.py` — Theorems A–D, algebraic consistency passed. Overall physical status: **[S]**.

Note on AI assistance

Prepared with support from Claude (Anthropic). Physical interpretations and conclusions: Johann Pascher (ORCID 0009-0000-6518-4064).

Bibliography

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[2] J. Pascher, *Doc. 012: Gravitational constant*, FFGFT corpus.

[3] J. Pascher, *Doc. 025: T0 Cosmology*, FFGFT corpus.

[4] J. Pascher, *Doc. 078: Time — exclusion theorem* $T \cdot m = 1$, FFGFT corpus.

[5] J. Pascher, *Doc. 306: Native time-energy reciprocity*, FFGFT corpus (R50).

[6] J. Pascher, *Doc. 049: Lagrangian comparison*, FFGFT corpus.

[7] J. Pascher, *Doc. 344: Furey fermions in GALG and FFGFT*, FFGFT corpus (Sept. 2026).

[8] J. Pascher, *Doc. 349:* $SU(2)_L$ *chirality from Galois*, FFGFT corpus (Sept. 2026).

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